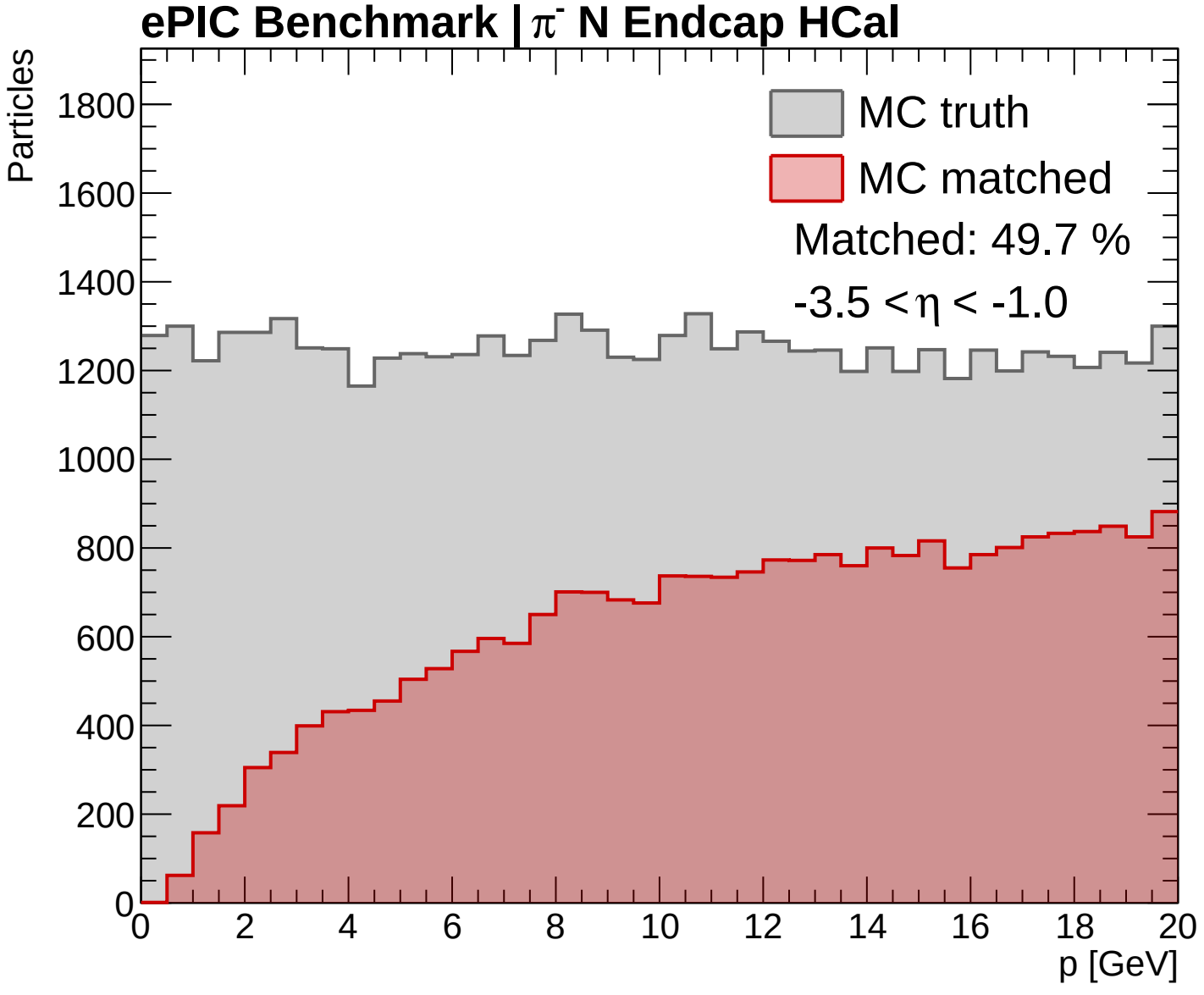
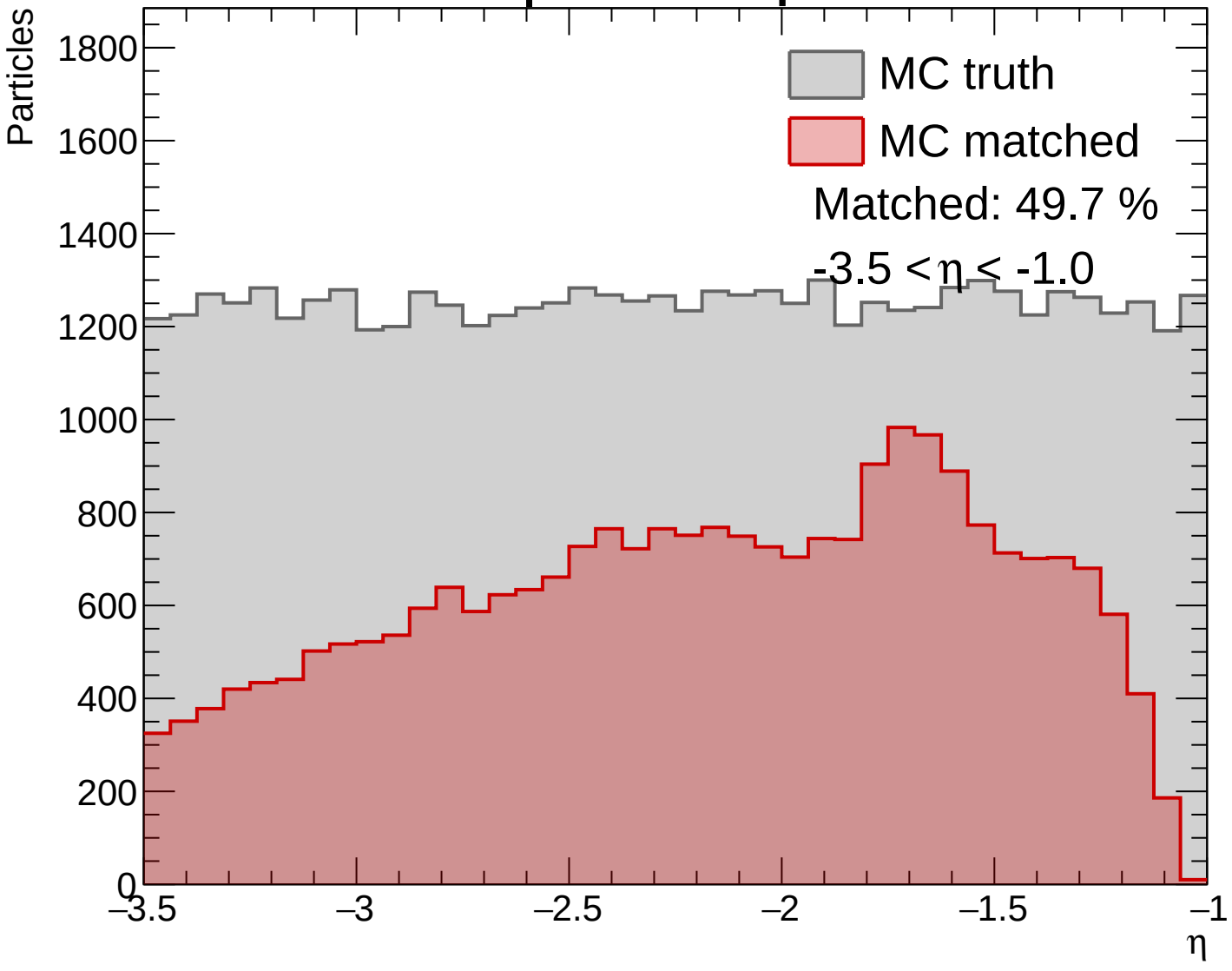
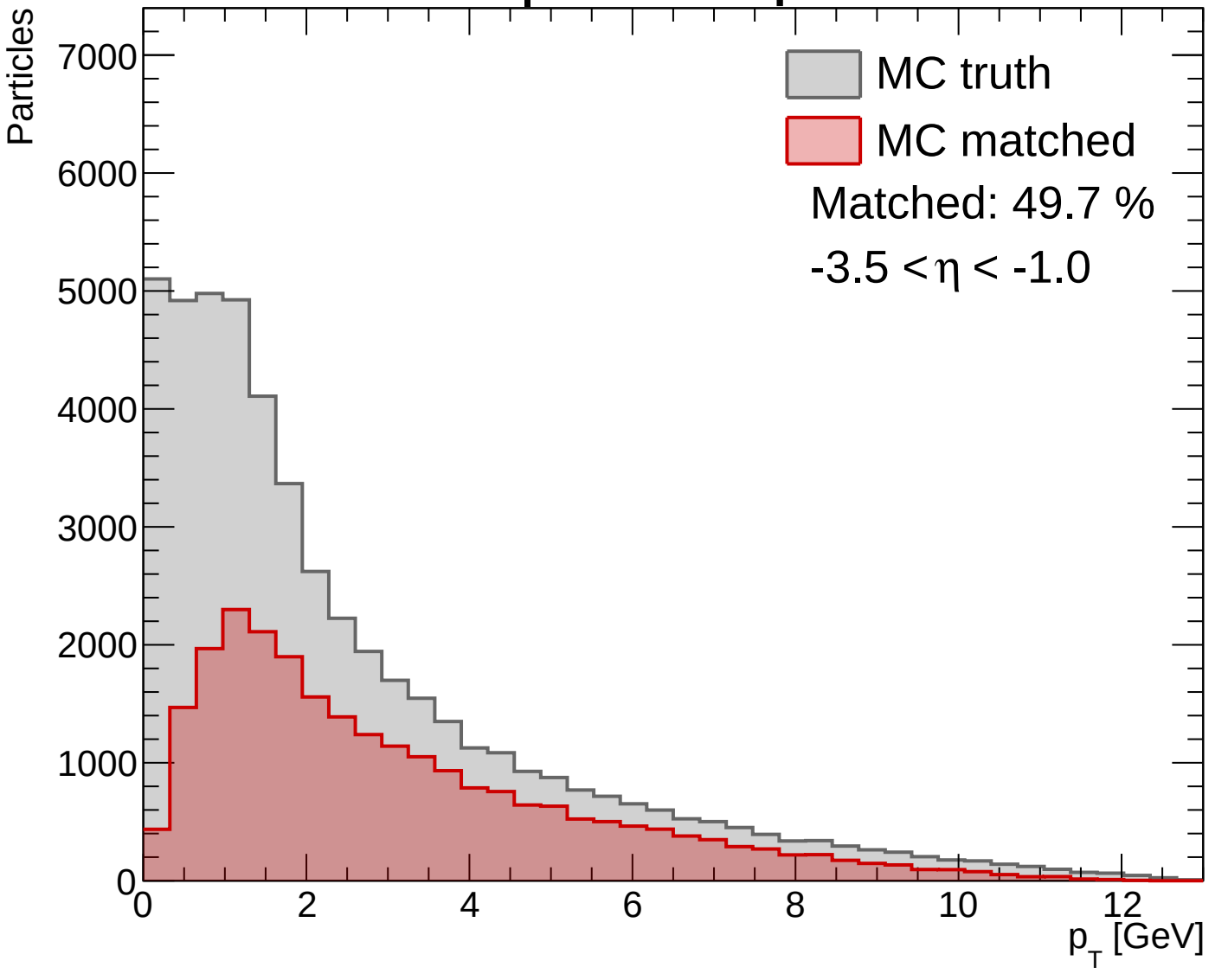
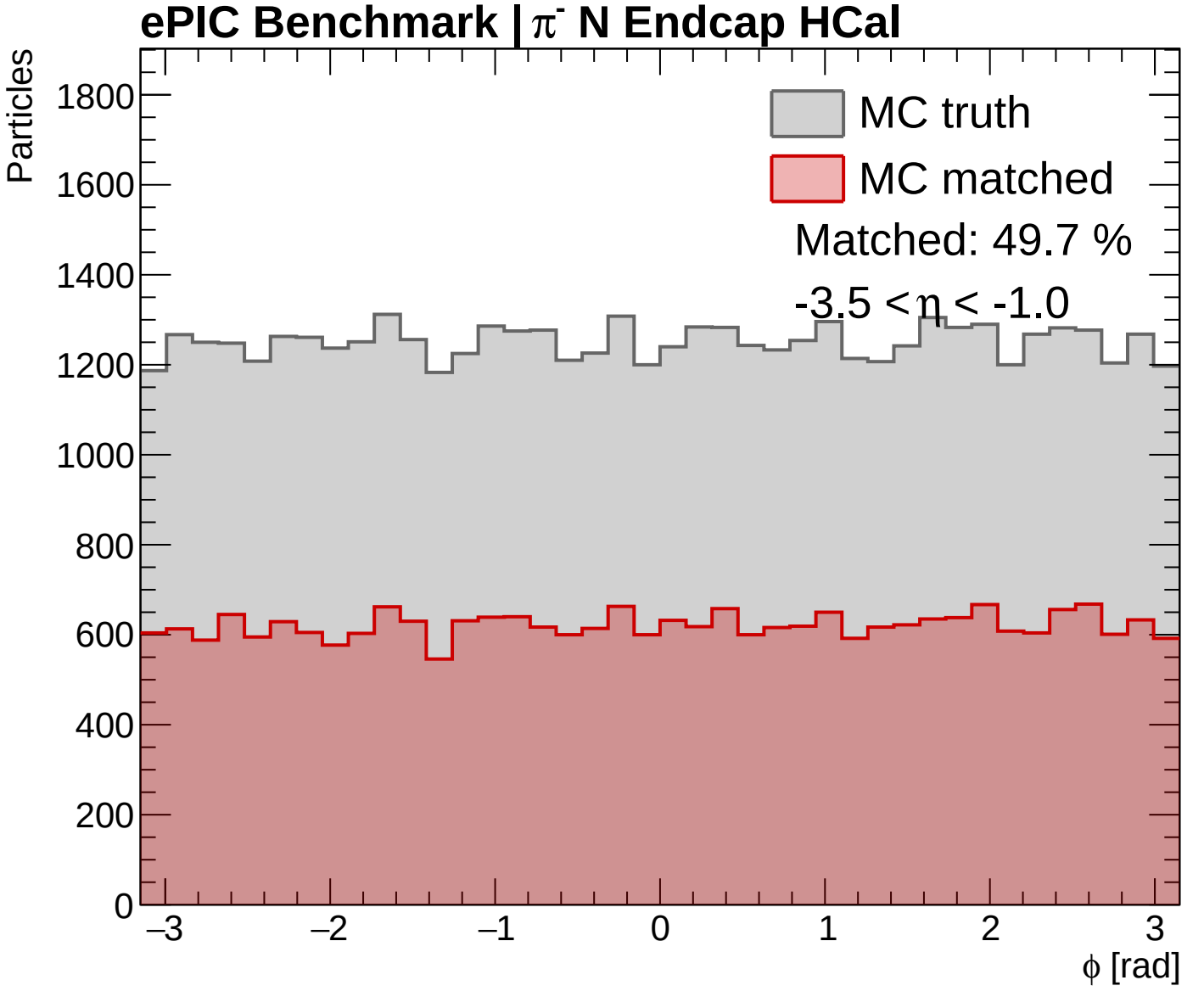
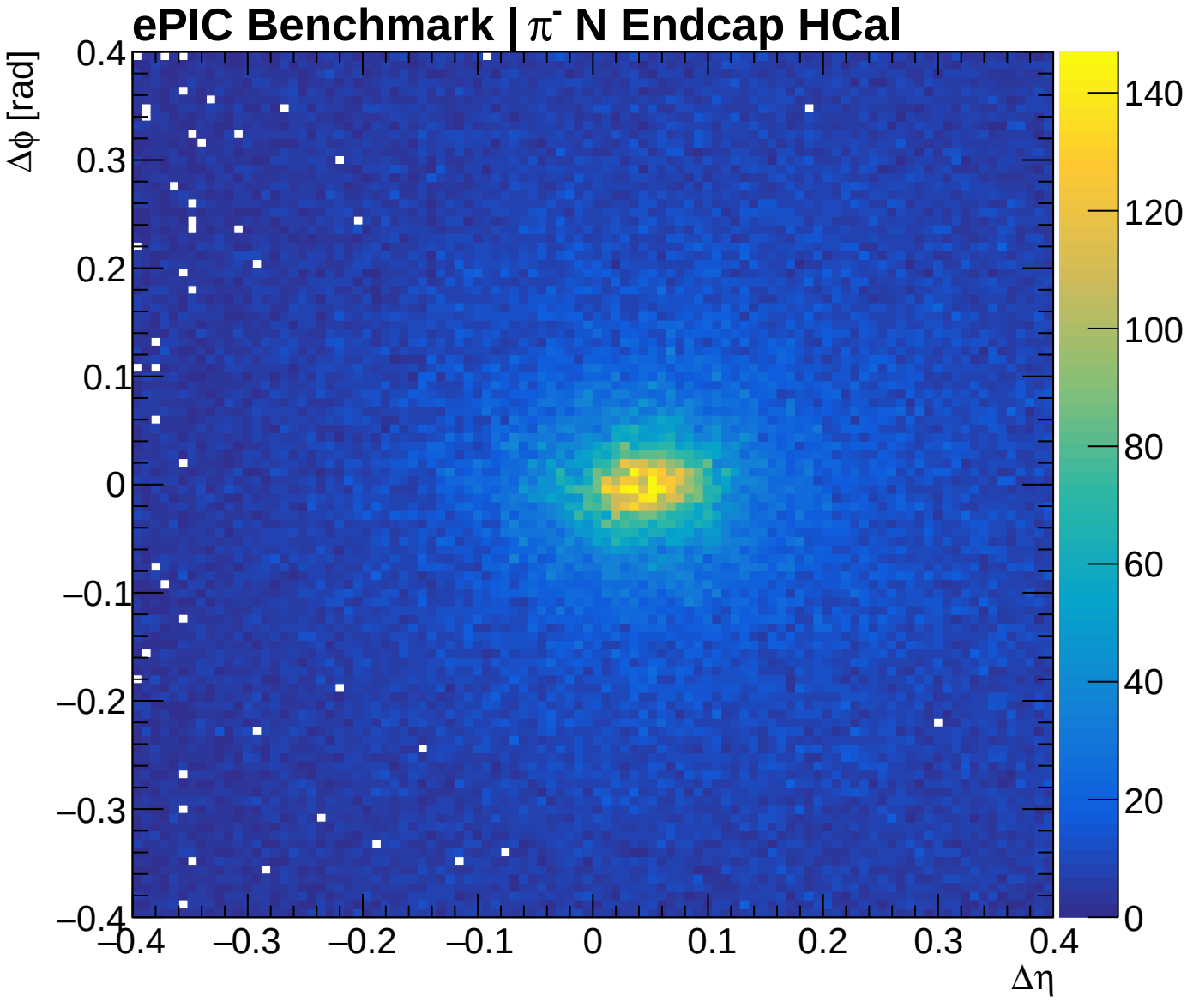
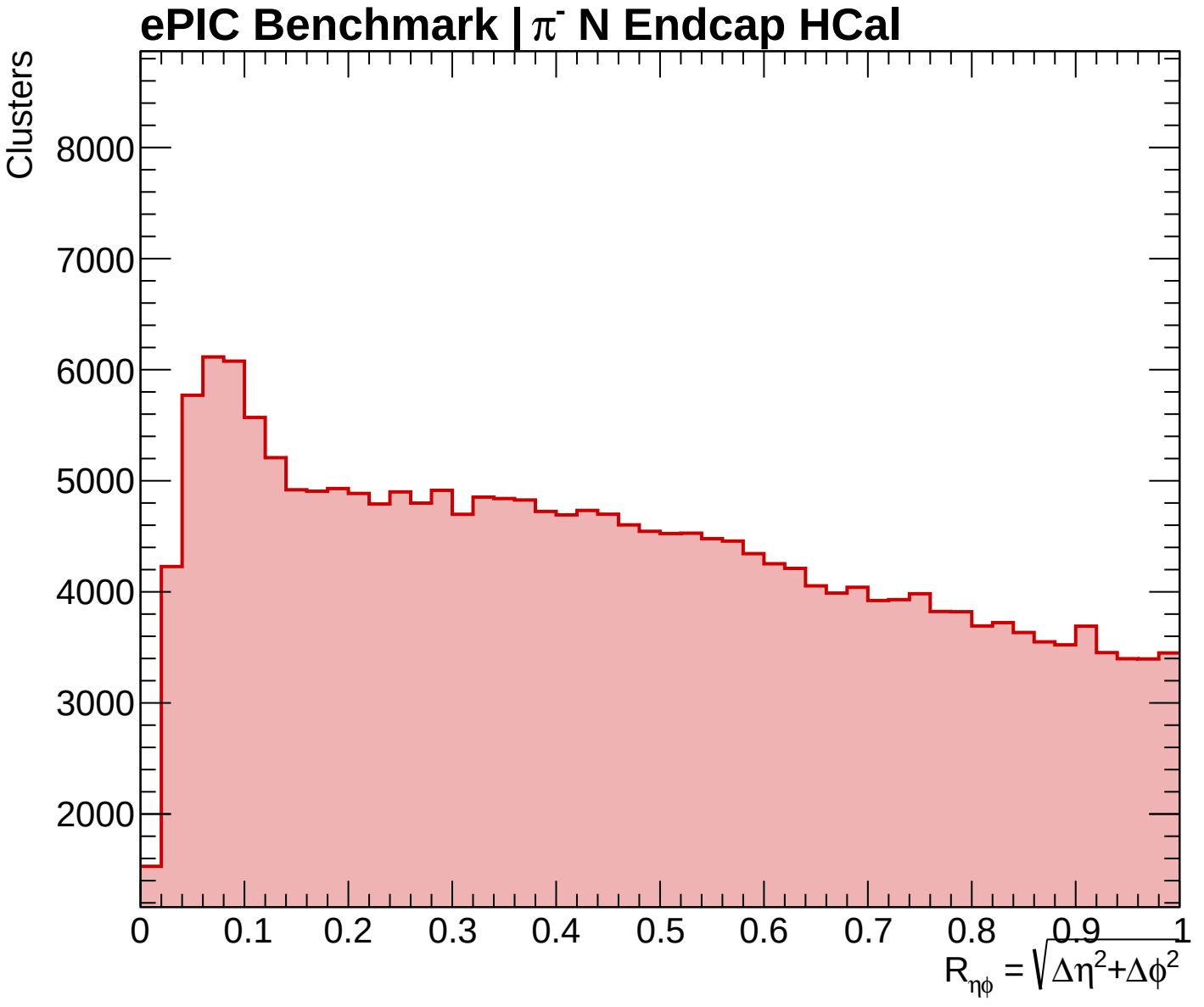

ePIC Benchmark | π⁻ N Endcap HCal
Particles
MC truth
MC matched
Matched: 49.7 %
-3.5 < η < -1.0
0
200
400
600
800
1000
1200
1400
1600
1800
0
2
4
6
8
10
12
14
16
18
20
p [GeV]

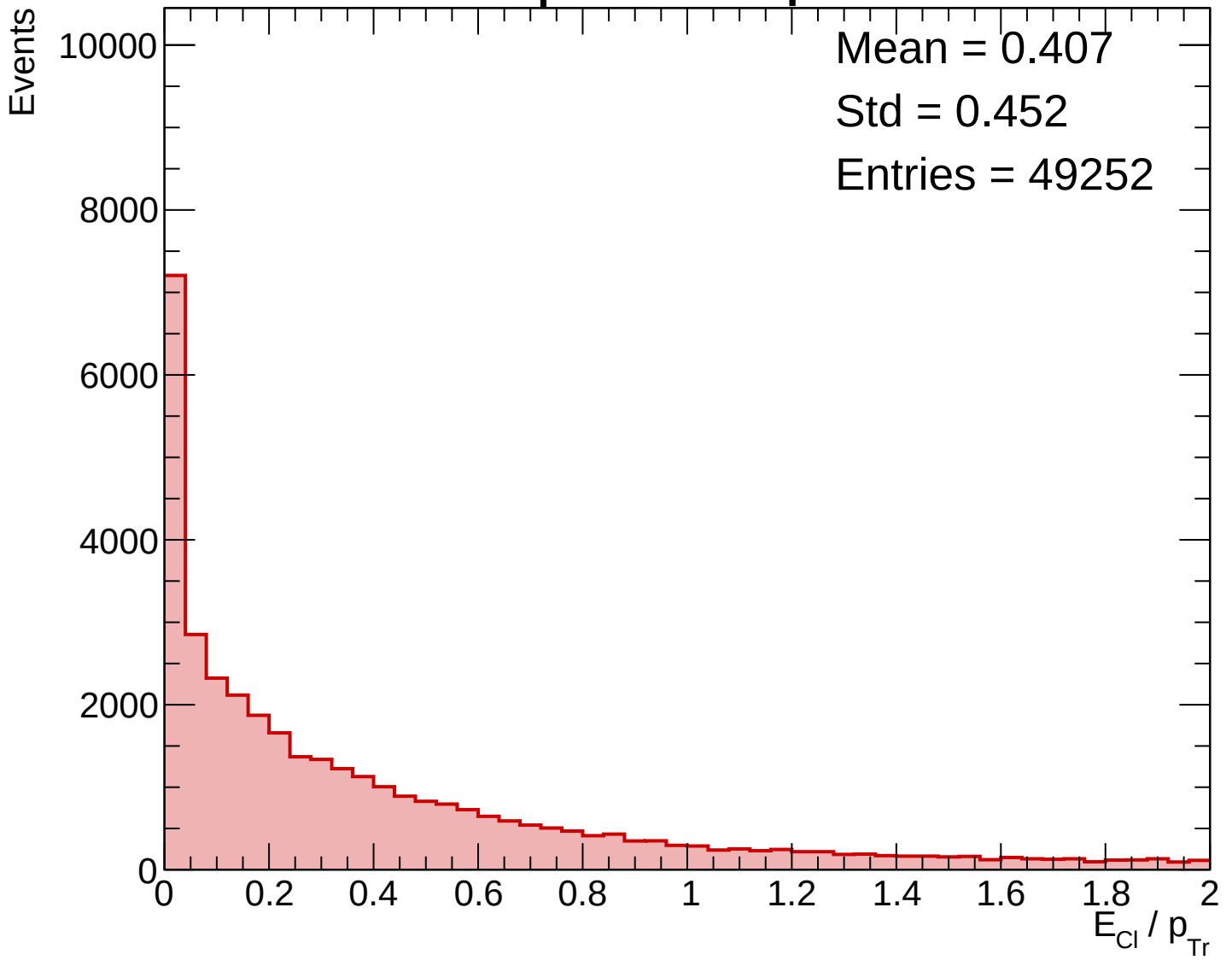

ePIC Benchmark | π⁻ N Endcap HCal
Particles
1800
1600
1400
1200
1000
800
600
400
200
0
MC truth
MC matched
Matched: 49.7 %
-3.5 < η < -1.0
–3.5
–3
–2.5
–2
–1.5
–1
η

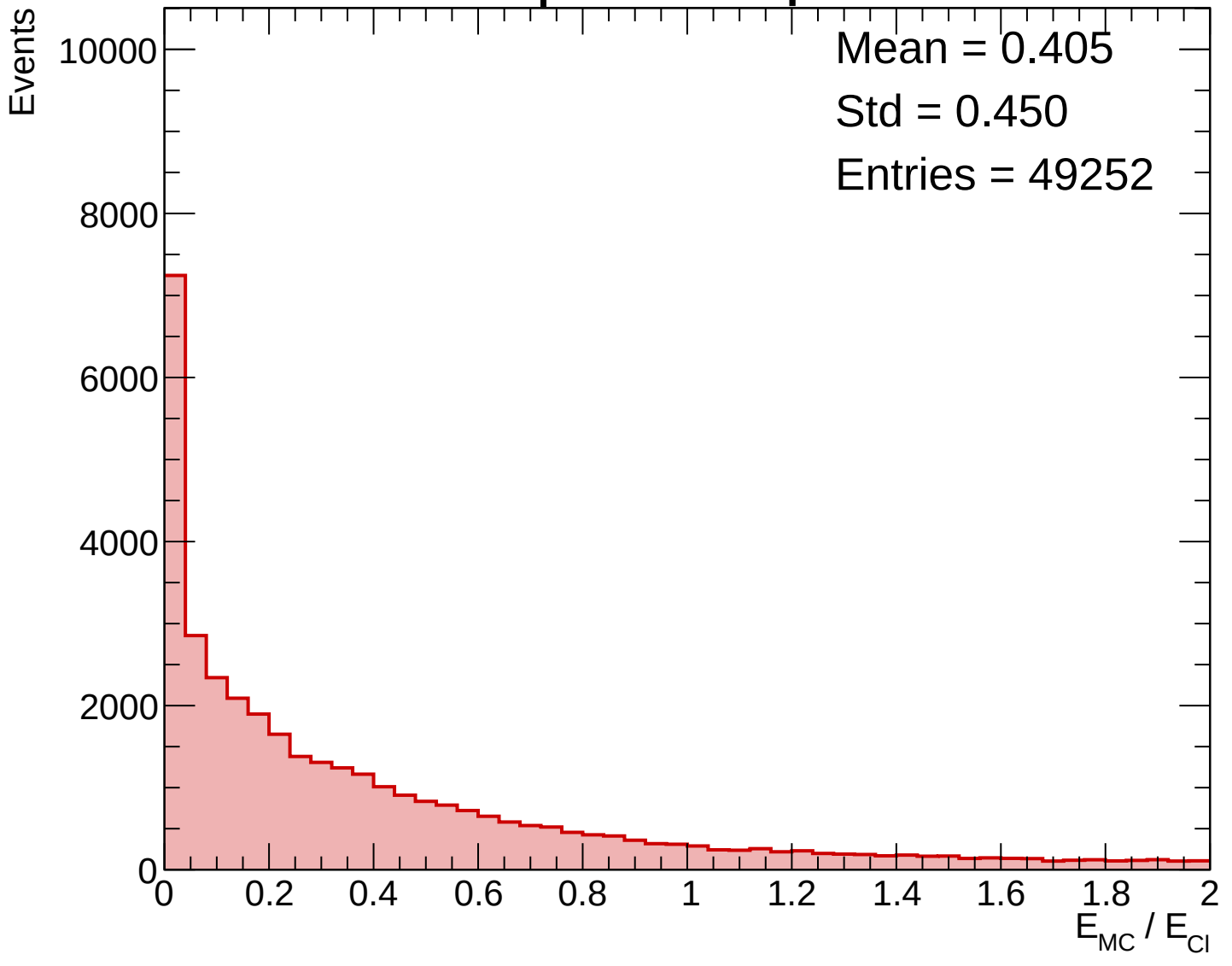

ePIC Benchmark | π⁻ N Endcap HCal
Particles
7000
6000
5000
4000
3000
2000
1000
0
0
2
4
6
8
10
12
p_T [GeV]
MC truth
MC matched
Matched: 49.7 %
-3.5 < η < -1.0

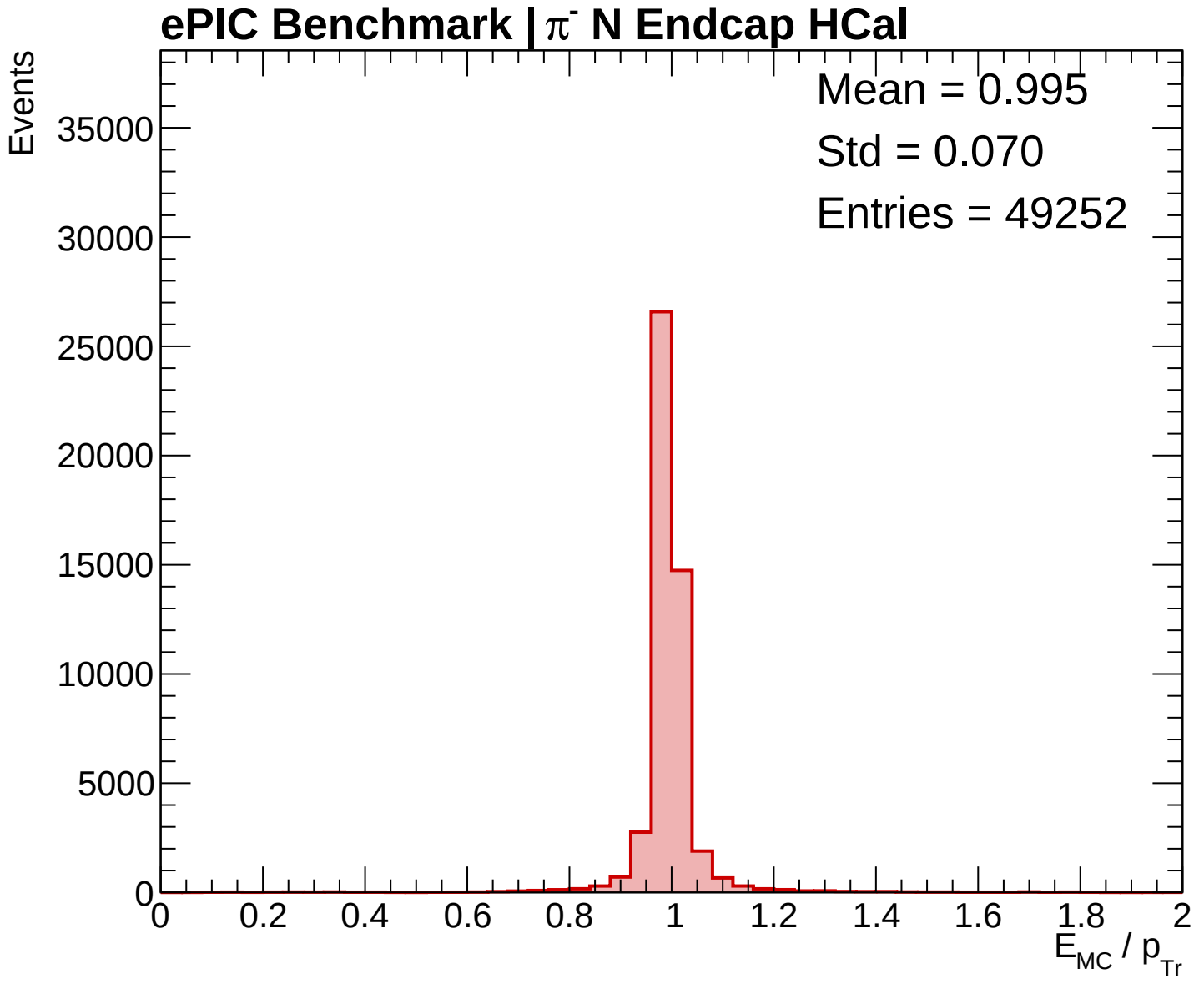

ePIC Benchmark | π⁻ N Endcap HCal
Particles
1800
1600
1400
1200
1000
800
600
400
200
0
MC truth
MC matched
Matched: 49.7 %
-3.5 < η < -1.0
–3
–2
–1
0
1
2
3
ϕ [rad]

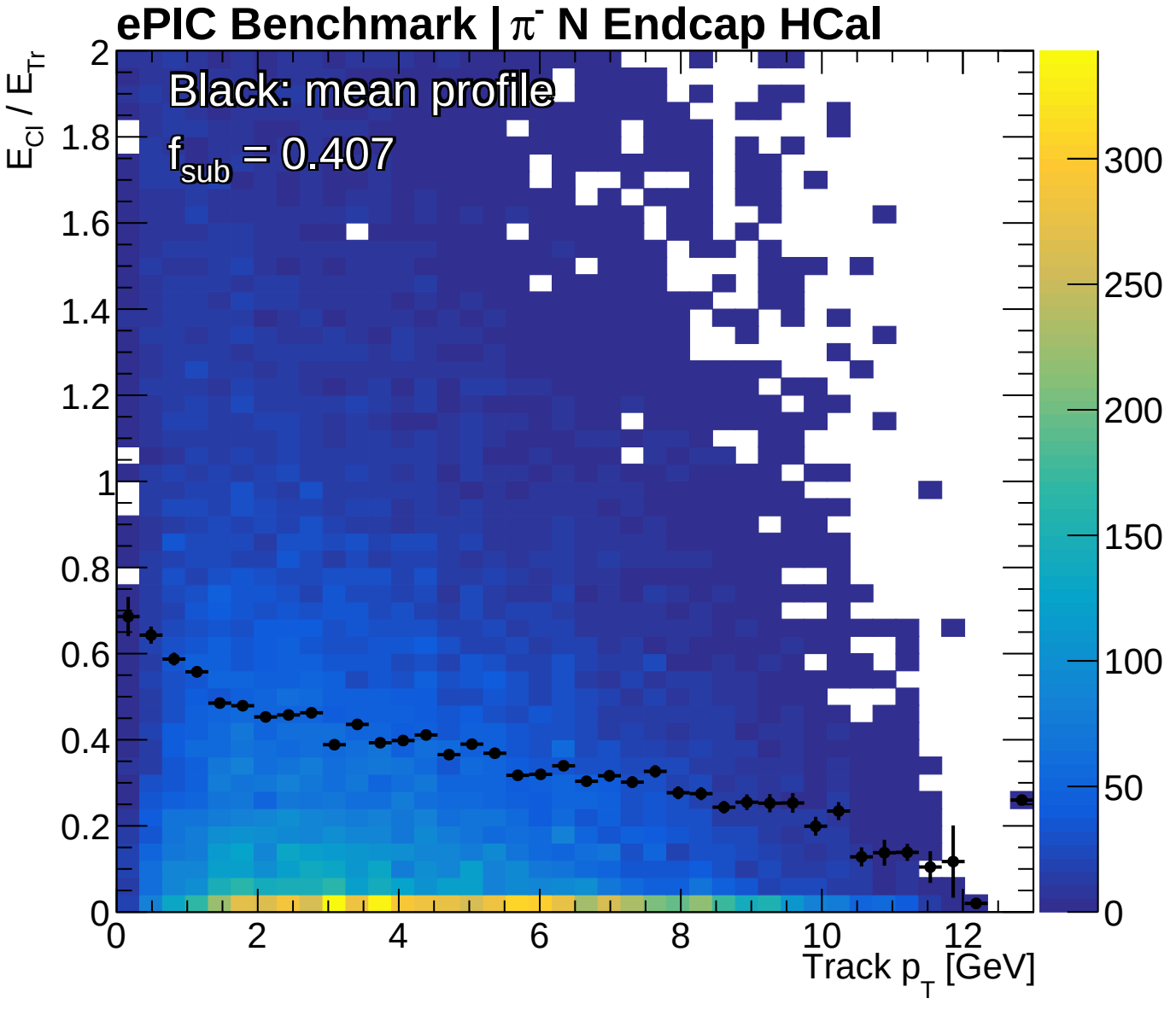

ePIC Benchmark | π⁻ N Endcap HCal
Δϕ [rad]
0.4
0.3
0.2
0.1
0
–0.1
–0.2
–0.3
–0.4
–0.4
–0.3
–0.2
–0.1
0
0.1
0.2
0.3
0.4
Δη
140
120
100
80
60
40
20
0

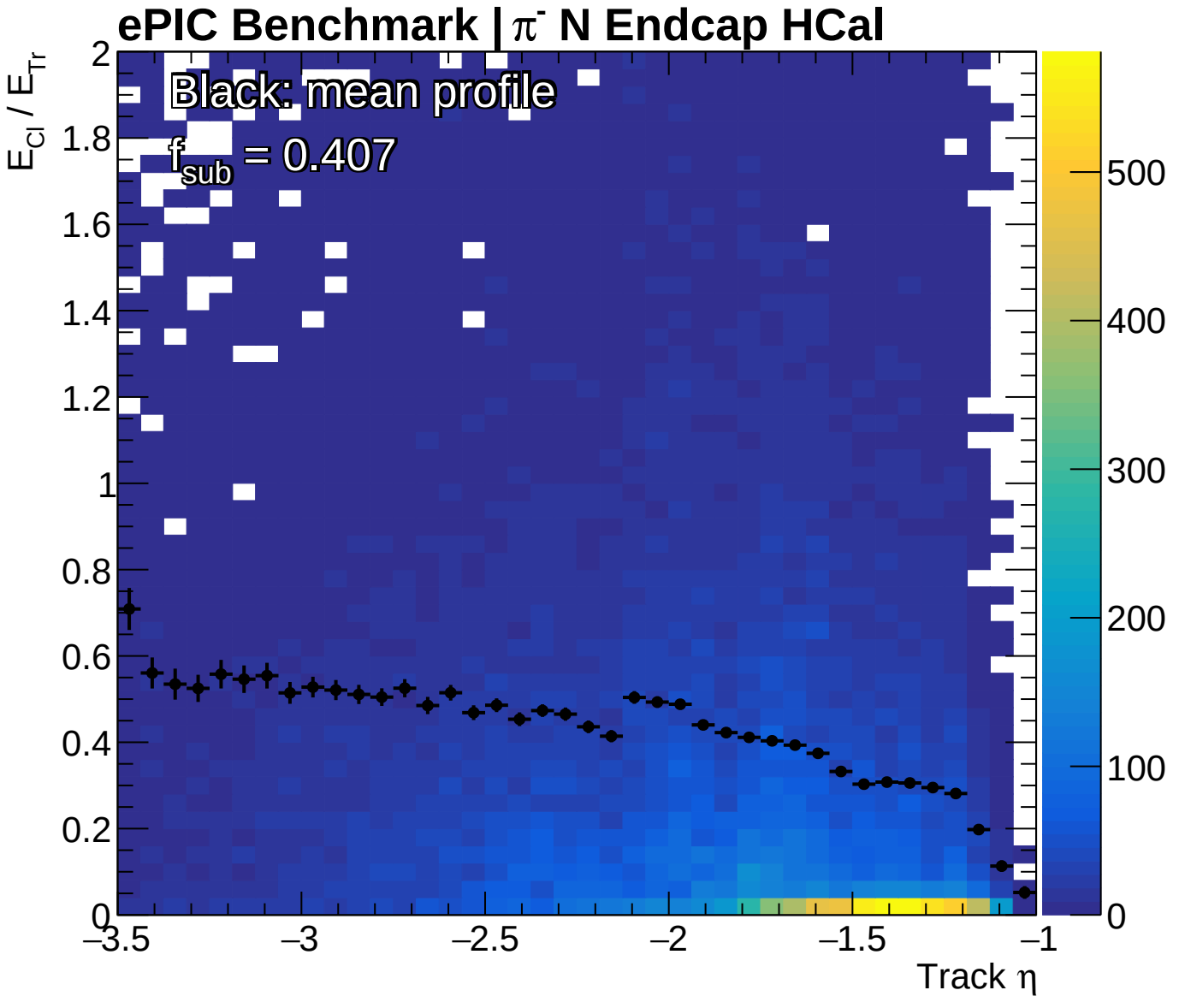

ePIC Benchmark | π⁻ N Endcap HCal
Clusters
8000
7000
6000
5000
4000
3000
2000
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
$R_{\eta\phi} = \sqrt{\Delta\eta^2+\Delta\phi^2}$

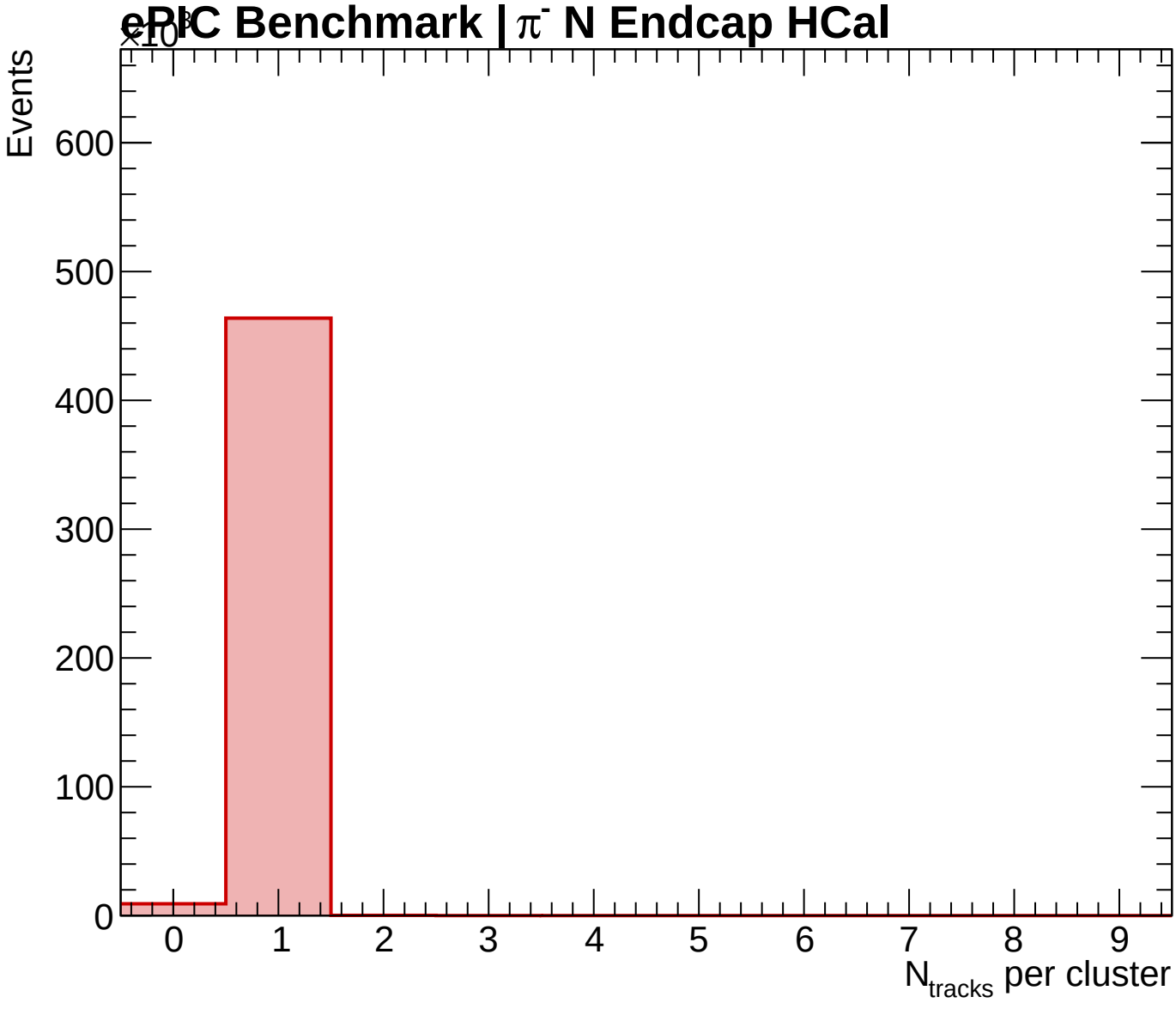

ePIC Benchmark | π⁻ N Endcap HCal
Mean = 0.407
Std = 0.452
Entries = 49252
Events
10000
8000
6000
4000
2000
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_Cl / p_Tr

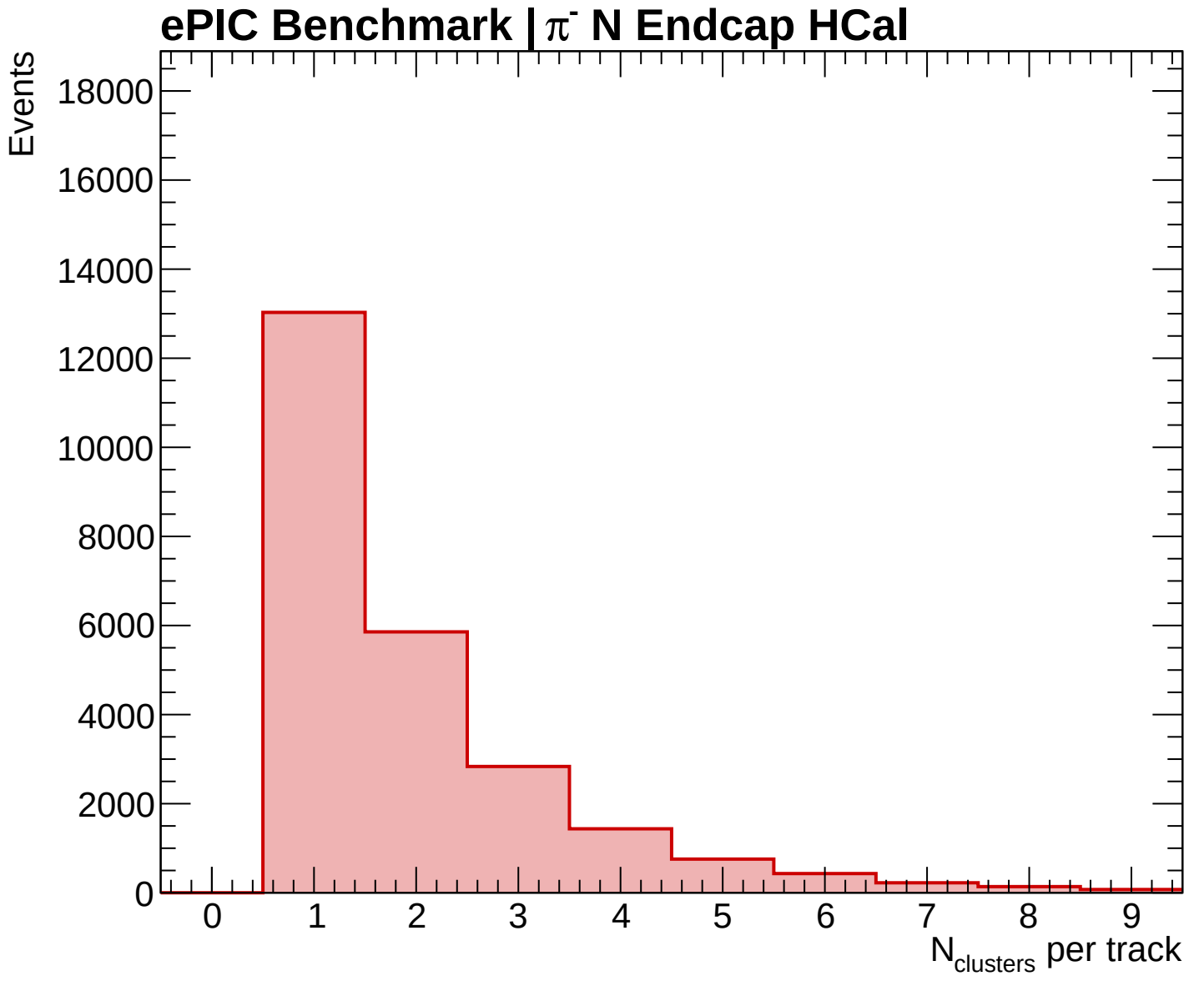

ePIC Benchmark | π⁻ N Endcap HCal
Mean = 0.405
Std = 0.450
Entries = 49252
Events
10000
8000
6000
4000
2000
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_MC / E_Cl

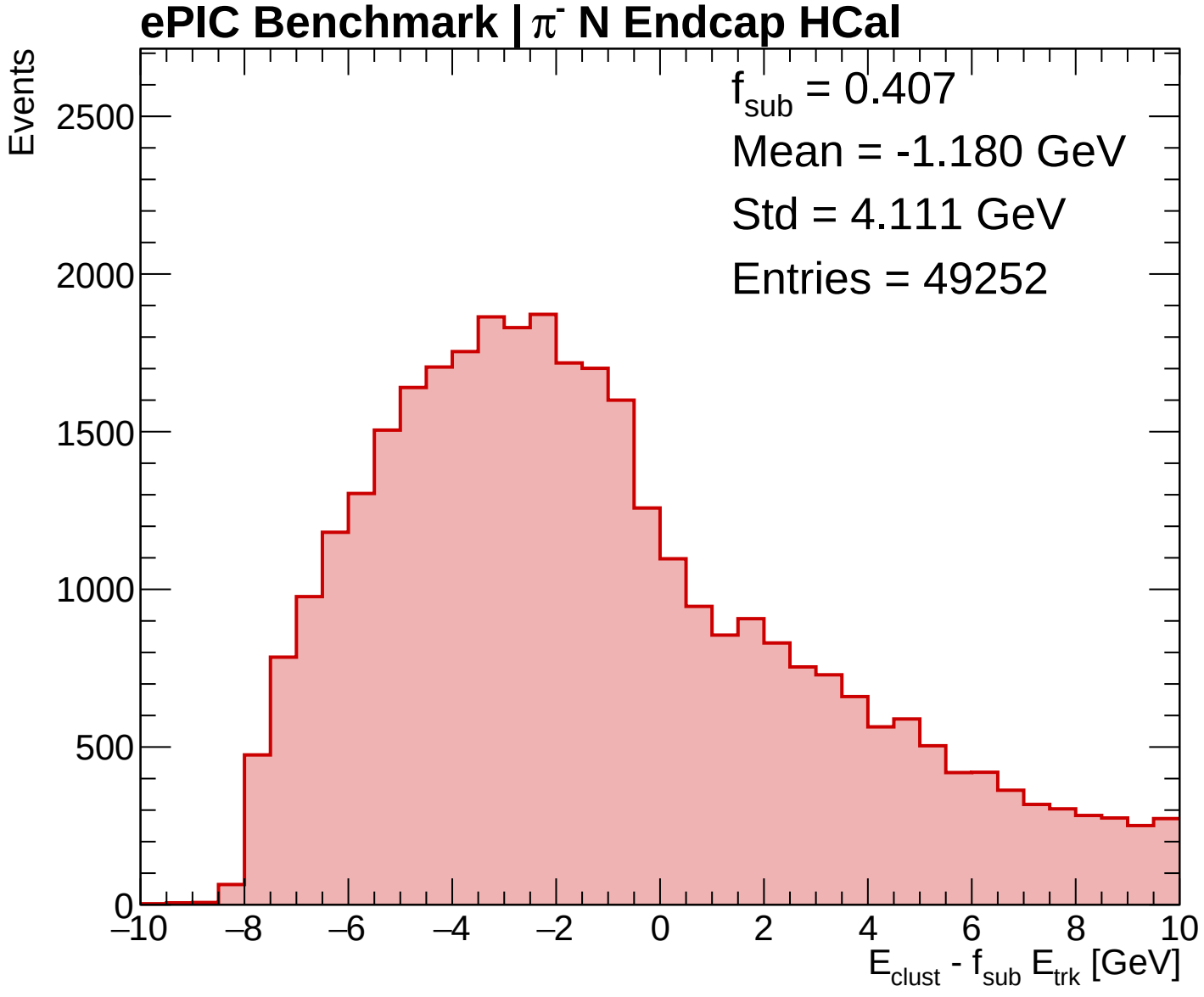

ePIC Benchmark | π⁻ N Endcap HCal
Mean = 0.995
Std = 0.070
Entries = 49252
Events
35000
30000
25000
20000
15000
10000
5000
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_MC / p_Tr

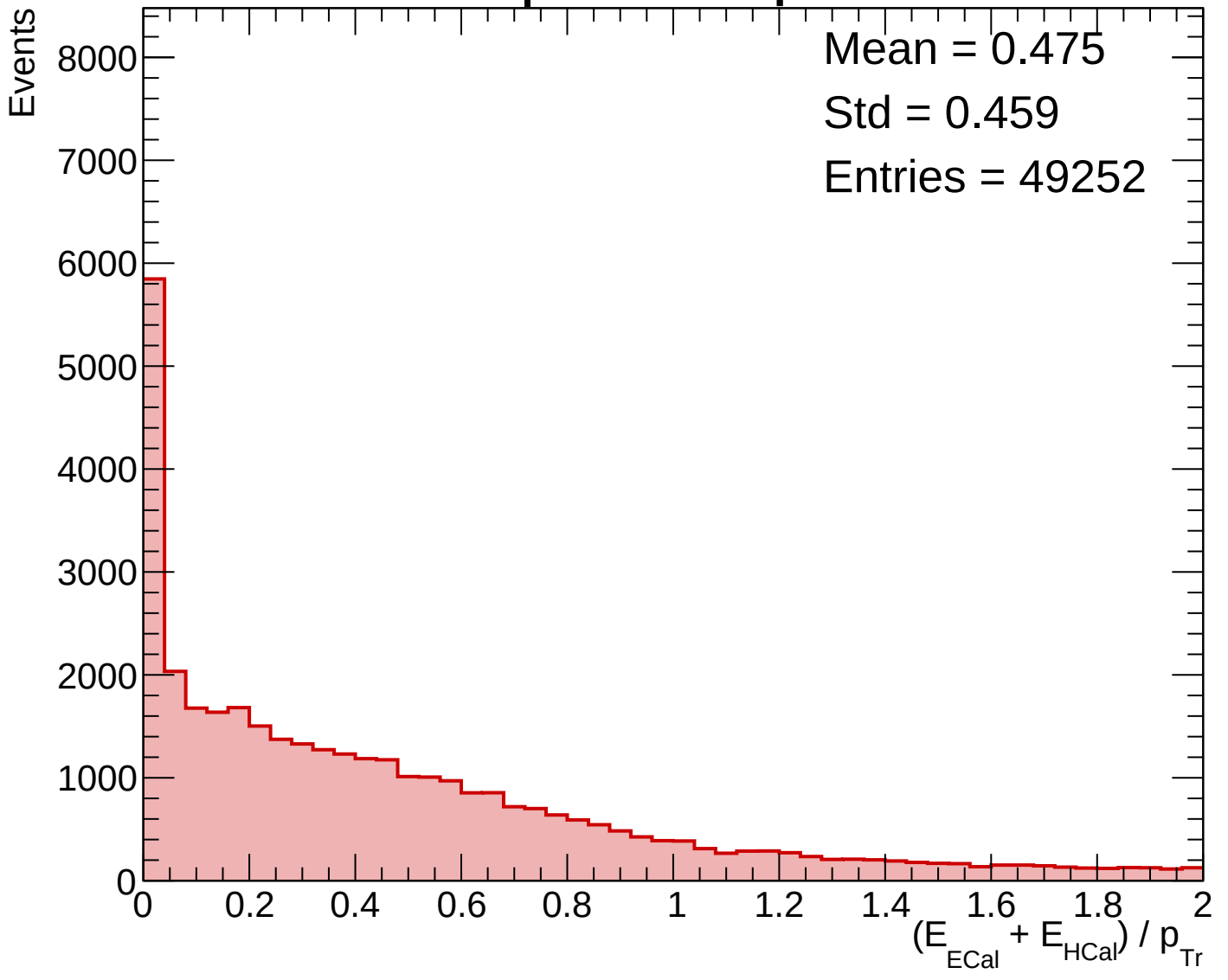

ePIC Benchmark | π⁻ N Endcap HCal
Black: mean profile
f_sub = 0.407
E_Cl / E_Tr
Track p_T [GeV]
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0
2
4
6
8
10
12
0
50
100
150
200
250
300


ePIC Benchmark | π⁻ N Endcap HCal
Black: mean profile
f_sub = 0.407
E_Cl / E_Tr
Track η
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
–3.5
–3
–2.5
–2
–1.5
–1
500
400
300
200
100
0


ePIC Benchmark | π⁻ N Endcap HCal
×10³
Events
600
500
400
300
200
100
0
0
1
2
3
4
5
6
7
8
9
N_tracks per cluster


ePIC Benchmark | π⁻ N Endcap HCal
Events
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
1
2
3
4
5
6
7
8
9
N_clusters per track


ePIC Benchmark | π⁻ N Endcap HCal
Events
2500
2000
1500
1000
500
0
f_sub = 0.407
Mean = -1.180 GeV
Std = 4.111 GeV
Entries = 49252
–10
–8
–6
–4
–2
0
2
4
6
8
10
E_clust - f_sub E_trk [GeV]


ePIC Benchmark | π⁻ N Endcap HCal
Mean = 0.475
Std = 0.459
Entries = 49252
Events
8000
7000
6000
5000
4000
3000
2000
1000
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
(E_ECal + E_HCal) / p_Tr